

when the value of $f(n)$ is fixed or constant, its time complexity is represented as a rule of 1 or big O of n

Time or Space Trade off

These are the fundamental concepts of algorithm analysis and design.

They refer to the idea that the programmer can often optimize an algorithm to improve its performance through time complexity & space complexity.

When optimizing algorithm when a programmer make a trade off between time or space optimization.

Time Optimization → When optimizing algorithm the programmer can reduce its time complexity

It may be more efficient or require more memory space

Space optimization
programmer can optimize an algorithm to reduce its space complexity
This can be done by using more compact data structure optimizers - auxiliary space

However it may increase the cost and time due to additional computation or storage data access